

Math 141, S10

Quiz 9

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1.(10 points) Find the following limits using the L'Hopital's Rule.

$$(a) \lim_{x \rightarrow +\infty} \frac{\ln x \rightarrow +\infty}{\sqrt{x} \rightarrow +\infty} = \lim_{x \rightarrow +\infty} \frac{(\ln x)'}{(\sqrt{x})'} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow +\infty} \frac{2}{\sqrt{x}} = 0$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos x \rightarrow 0}{x^2 \rightarrow 0} = \lim_{x \rightarrow 0} \frac{(1 - \cos x)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} \\ = \lim_{x \rightarrow 0} \frac{(\sin x)'}{(2x)'} = \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2}$$

$$(c) \lim_{x \rightarrow 0^+} x^x$$

$$\text{Let } y = \ln(x^x) = x \ln x. \\ \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{(\frac{1}{x})} = \lim_{x \rightarrow 0^+} \frac{(\frac{1}{x})'}{(-\frac{1}{x^2})} = \lim_{x \rightarrow 0^+} (-x) = 0 \\ \Rightarrow \lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^y = e^0 = 1.$$